MILITARY SPECIFICATION

PELLETS, PLASTIC BONDED HMX (95/5) FOR USE IN AMMUNITION

1. SCOPE

1.1 This specification covers explosive pellets designated as Plastic Bonded HMX (PBHMX). (see 6.4)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the isssue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-T-45	-	Tape, Gummed Paper Reinforced and Plain for Sealing and Securing
PPP-B-636	-	Box, Fiberboard

MILITARY

MIL-D-1000	-	Drawings, Engineering and Associated Lists
MIL-B-2427	-	Boxes, Ammunition Packing: Wood, Nailed
MIL-L-10547	-	Liners, Case Waterproof
MIL-H-45444	-	HMX
MIL-P-50854	-	Plastic Bonded HMX (95/5) Powder

STANDARDS

FEDERAL

FED-STD-101	-	Preservation, Packaging, and Packing Materials: Test Procedures.
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1376

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MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-650 - Explosive: Sampling Inspection and Testing
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1171 - Acceptance and Description Sheets for Propellants and Explosives
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

8796522 - Marking Diagram and Sealing for Wood Packing Boxes
P9209048 - Pellet, H.E.

PUBLICATIONS

ET-P9209048 - Equipment Tabulation

(Copies of specification, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other Publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM D 792-66 - Specific Gravity and Density of Plastics by Displacement

(Requests for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania 19103.)

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3. REQUIREMENTS

3.1 Material.-The PBHMX pellets shall conform to all the requirements specified on drawing P9209048 and the specifications referenced thereon.

3.2 Composition.-The PBHMX pellets shall be homogeneous. The requirements for chemical composition of the PBHMX pellets shall be as specified in MIL-P-50854 (MU) when determined as specified in 4.4.1.

3.3 Density.-The density of the pellets shall conform to the requirement specified on drawing P9209048 when tested as specified in 4.4.2.

3.4 Moisture.-The moisture content shall conform to the requirement specified on drawing P9209048 when tested as specified in 4.4.3.

3.5 Moisture content of packaging material.-The moisture content of the paperboard products (fiberboard boxes, separators and fillers) and wood products (sawdust) shall be 9.0 percent, maximum, when determined as specified in 4.4.4.

3.6 Workmanship.-The pellets and packaging material shall be free of dirt, grease and other foreign matter. The pellets shall be firmly compacted, free of soft spots and powdery areas.

3.7 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

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a. A statement that the lot complies with all requirements and quality assurance provisions specified in this specification.

b. Specification number and date, together with an identification and date of changes.

c. Certificates of analysis on all materials used directly by the contractor when such material is controlled by Government specifications shall be made available upon request by the Contracting Officer.

d. Quantity of product in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principle. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample consisting of 10 PBHMX pellets in accordance with instructions issued by the Contracting Officer for evaluation in accordance with paragraph 4.2.2. All samples submitted shall have been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of analysis. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer.

4.2.2 Inspections to be performed.-The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.3 and 4.4 of this specification and any or all requirements of the applicable drawings.

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4.2.3 Rejection.-If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.-The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Each lot of PBHMX pellets shall be produced from one lot of PBHMX powder produced in accordance with MIL-P-50854 (MU). The product shall be submitted for inspection in accordance with MIL-STD-105. The criteria and procedure for the assignment of lot numbers shall be in accordance with Standard MIL-STD-1168.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with MIL-STD-105 (ABC-STD-105). Contractor's sampling plans, if used, shall be approved by the Government and shall provide as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed, using an AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each minor defect.

4.3.2.1 PBHMX pellets (see dwg. P9209048)

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		

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Major:	AQL 1.0 Percent		
101.	Weight	Balance	01001
102.	Length	Gage	01002
103.	Taper on outside surface	Gage	01003
104.	Taper on inside surface	Gage	01004
105.	Depth of cavity	Gage	01005
106.	Concentricity of diameter of cavity with respect to outside diameter	Gage	01006
107.	Crack or lamination	Visual	01007
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	01008

4.3.2.2 Carton prior to closing

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 Percent		
101.	Separator improperly assembled or damaged	Visual/ Manual	02001
Minor:	AQL 0.65 Percent		
201.	Foreign matter	Visual	02002

4.3.2.3 Carton, Sealed

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Contents loose	Manual	03001
102.	Explosive content overweight	Balance	03002
103.	Stapling or stitching improper or insecure	Visual	03003
Minor:	AQL 1.50 Percent		
201.	Carton closure insecure (see 5.1.1)	Visual/ Manual	°03004
202.	Marking misleading or unidentifiable	Visual	03005
203.	Evidence of poor workmanship (see 3.6)	Visual	03006

4.3.2.4 Box, wood packing, prior to closing

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		

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101.	Splinters or foreign matter.....	Visual	04001
102.	Liner punctured or torn.....	Visual	04002
103.	Liner improperly closed.....	Visual	04003
Minor:	AQL 0.65 Percent		
201.	Type of liner incorrect.....	Visual	04004

4.3.2.5 Box, wood packing, sealed

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 Percent		
101.	Board broken or split.....	Visual	05001
102.	Contents exposed.....	Visual	05002
103.	Strapping misssing, broken or loose..	Visual/ Manual	05003
Minor:	AQL 1.50 Percent		
201.	Contents loose.....	Manual	05004
202.	Strapping improperly assembled.....	Visual	05005
203.	Marking misleading or unidentifiable.	Visual	05006

4.3.3 Sampling.-A minimum of ten (10) pellets shall be removed from the beginning of the pelletizing operation and a minimum of 10 pellets shall be removed from the end of the pelletizing operation. The two 10 pellet samples shall be crushed and retained as two separate individual samples. These two samples shall be tested for the composition and moisture requirement in accordance with the tests depicted in 4.4.1 and 4.4.3. If either of these samples fail to comply with the requirements specified in 3.2 and 3.4 the lot of PBHMX pellets shall be rejected. If the total weight of either of the samples described above (beginning or end) is less than 100 grams, then additional pellets shall be selected from the pelletizing operation (beginning and end) to obtain a minimum weight of 100 grams for each sample.

4.3.3.1 Density.-A random sample of ten (10) individual pellets shall be selected from the sample used in 4.3.2.1. If the density of these pellets fails to meet the requirement specified in 3.3, the lot shall be rejected.

4.3.3.2 Sampling for Moisture Content of Packaging Materials.- Paperboard products (fiberboard boxes, separators and fillers) and wood products (sawdust) shall be sampled in accordance with Special Inspection Level S-3 with AQL 4.0 percent. The sample size represents the number of each of the above components to be sampled except that the sample size of sawdust represents the number of individual containers (bags) to be sampled.

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4.3.4 Inspection equipment.-Equipment Tabulation (ET) Number ET-P9209048 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test Methods and Procedures

4.4.1 Composition

4.4.1.1 Plastic binder, Major Defect, Defect Code. No. 06001.- Transfer an accurately weighed portion of approximately 1 gram of the sample to a tared medium porosity fritted glass crucible. Extract the contents of the crucible with 4, 25 ml portions of toluene saturated with HMX. Stir the contents of the crucible after the addition of each portion and allow the solvent to remain in contact with the sample for 3 minutes before applying suction. Aspirate the crucible and contents until the odor of toluene is no longer perceptible, dry in an oven maintained at 100 $\pm$ 5 C for 1 hour, cool in a desiccator and weigh. Calculate the loss in weight of the crucible and contents to percentage plastic binder as follows:

$$\text{Percent Plastic binder} = \frac{100\ A}{B}$$

Where:

A = loss in weight, in g
B = wt. of sample on dry basis, in g

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4.4.1.2 HMX content, Major Defect, Defect Code No. 07001.-The percent of HMX in the sample shall be calculated by subtracting from 100 the percent plastic binder obtained as specified in 4.4.1.1.

4.4.2 Density, Major Defect, Defect Code No. 08001.-The density shall be determined in accordance with ASTM Method D 792-66, Method A-1. Calculate the density of the PBHMX pellets as follows (see 6.5):

$$\text{Density, minimum} < \overline{X} - 3s$$

Where:

$\overline{X}$ = average of ten (10) individual pellets
s = standard deviation

4.4.3 Moisture, Major Defect, Defect Code No. 09001.-The moisture shall be determined in accordance with Standard MIL-STD-650, Method 101.4, using methanol as the special solvent.

4.4.4 Moisture content of packaging material.

4.4.4.1 Moisture content of paperboard products, Major Defect, Defect Code No. 10001.-The moisture content of the paperboard products (fiberboard boxes, separators and fillers) shall be determined in accordance with Federal Test Method Standard No. 101, Method 5024, using the ovendrying method.

4.4.4.2 Moisture content of wood products, Major Defect, Defect Code No. 11001.-The moisture content of the wood products (sawdust) shall be determined in accordance with Federal Test Method Standard No. 101, Method 5025, using the ovendrying method.

5. PREPARATION OF DELIVERY

5.1 Packaging

5.1.1 Level C.-Two hundred (200) pellets shall be packaged in a Type CF, Style RSC, Class Domestic, Variety SW, "B" Flute, Grade 175 or 200 fiberboard box meeting the requirements of Specification PPP-B-636. Individual cells shall be provided by half-slotted partitions assembled in a 10 by 5 by 4 layer per box arrangement. A small filler shall be used between top and bottom flaps of the box and a large filler shall be used between each layer of pellets. A large filler shall be placed on top of the contents and additional large fillers may be added as required to prevent separator motion. Separators and fillers shall be made from the same material as that used for box constructions. The box size shall be approximately 14 1/4 by 7 1/8 by 6 5/8 inches deep inside dimensions. The separator size shall be a slide fit in length and width of fiberboard box by 1 1/2 inches high. Cell size shall be approximately 1 5/16 by 1 5/16 inches. Pellets shall be placed in cells with small diameter down. Small filler size shall be approximately 7 X 7 inches. Large filler size

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shall be approximately 14 by 6 7/8 inches. The boxes shall be closed by applying 2 inch wide gummed kraft tape over entire length of all seams with approximately 2 inch extension at the end panels and around corners. Tape shall meet the requirements for Type III, Grade A or B of Specification PPP-T-45. All packaging material shall be thoroughly dry with moisture content not to exceed 9%.

5.2 Packing

5.2.1 Level C.-One fiberboard box shall be placed in a Type I, Class 1, Grade C wood box of specification MIL-B-2427 with a two inch cushion of clean, dry sawdust surrounding the fiberboard box. The wood box shall contain a liner conforming to Type I, Grade A of specification MIL-L-10547. Closure of the liner shall be in accordance with instructions given in Appendix of the liner specification. The inside box dimensions shall be approximately 16 3/4 by 9 5/8 by 9 3/8 inches deep. A commercial chipboard filler shall be placed on top of contents. Additional fillers, as required, shall be placed on top to provide a tight pack. Filler size shall be approximately 16 1/2 by 9 1/4 by 1/16 inches thick. The box shall be closed and strapped in accordance with the Appendix of the box specification.

5.3 Marking

5.3.1 Each fiberboard box shall be printed, stamped or labeled with black waterproof ink as follows:

200 PELLETS, H.E. P/N 9209048

LOT NO
DATE MO/YR

Size of letters and numbers shall be minimum 1/4 inches high and letters shall all be capitals. Wood box shall be marked in accordance with drawing 8796522. DOT marking shall be "HIGH EXPLOSIVES - DANGEROUS". Precautionary marking "THIS SIDE UP" shall appear in top of wood box in minimum 7/16 inches high capital letters.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

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b. Description sheets shall be prepared for each lot in accordance with Standard MIL-STD-1171.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs and inspection data submitted by the contractor.

6.3 Inspection equipment.-The contractor shall design and submit for approval inspection equipment as required by the referenced Equipment Tabulation (ET) and each applicable Equipment List (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.6.

6.3.1 Equipment Tabulation.-This tabulation (SMUPA Form 1011) is an index of all applicable EL's.

6.3.2 Equipment Lists (EL).-This list (SMUPA Form 1010) indicates the availability of inspection equipment designs by showing in the "Part or Identifying No." column, the numbers of drawings of Federal Stock Numbers of existing equipment designs, or codes as indicated in paragraph 6.3.3. Design action required of the contractor is described in paragraphs 6.3.4 and 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.5 for all the equipment coded as "Contractor Design in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6.

6.3.3 Equipment List Codes.-The inspection equipment as defined in 6.3.4 and 6.3.5 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army designs, mandatory for use.

6.3.4 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two

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basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.5) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6.

6.3.5 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three (3) copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.6.

6.3.6 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND8. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed to the Contracting Officer. Partial submission of inspection equipment designs

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is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission for designs. Picatinny Arsenal design review will be accomplished normally, within one month after receipt.

6.4 Intended Use.-The PBHMX pellets covered in this specification are intended to be used as the main charge of the XM41 grenade.

6.5 "Engineering Design Handbook", AMC Pamphlet 706-110 has been found suitable for determining the average and standard deviation of the PBHMX pellets.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1376-A-067

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

a. GIVE PARAGRAPH NUMBER AND WORDING.

b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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